

1. If E , M , J and G denote energy, mass, angular momentum and gravitational constant, then which of the following can have the dimensional formula of

$$\frac{EJ^2}{M^5G^2}?$$

- A) torque B) Mass
C) Planck's constant D) Angle
2. The gradient of a scalar function will be:
A) Normal to the surface
B) Tangential to the surface
C) Neither tangential nor normal to the surface
D) A scalar function
3. Eigen value of the matrix $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ is:
A) $e^{\pm \theta}$ B) $e^{\pm 2\theta}$ C) $e^{\pm i\theta}$ D) $e^{\pm i2\theta}$
4. The sequence represented by $\left(\frac{n}{e^n}\right)^\infty$ is:
A) Oscillating B) Monotonic increasing
C) Divergent D) Convergent
5. The function $f = \frac{z}{\sin z}$ at $z = \infty$ have:
A) No singularity B) Non isolated singularity
C) pole D) essential singularity
6. General solution for the differential equation $(D^2 + \omega^2)y = 0$ is:
A) $c_1 \cos \omega x + c_2 \sin \omega x$ B) $c_1 \cos \sqrt{\omega x} + c_2 \sin \sqrt{\omega x}$
C) $c_1 \sqrt{\cos \omega x} + c_2 \sqrt{\sin \omega x}$ D) $c_1 \cos^2 \omega x + c_2 \sin^2 \omega x$
7. In an n -dimensional space, the number of independent component that an antisymmetric tensor of rank 2 can have is :
A) n B) $2n$ C) $\frac{1}{2}n(n-1)$ D) $n(n+1)$

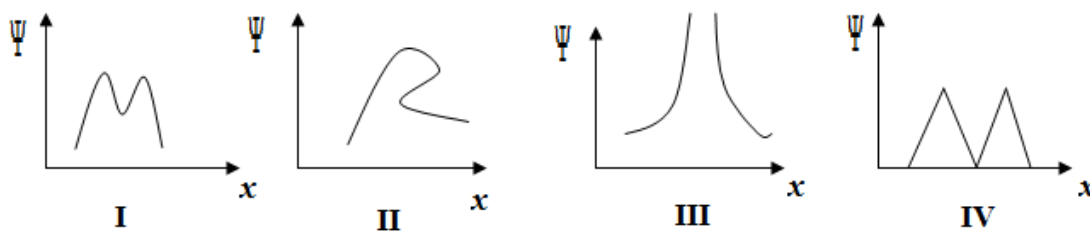
8. If $v = 3x^2y - y^3$ is a harmonic function, then the conjugate function corresponding to v is:
- A) $u = 3xy - y^2$ B) $u = x^3 - 3y^2x$
 C) $u = 3x^2y + y^3$ D) $u = 3x^3y + y^4$
9. The integral $\oint_C \frac{dz}{z(z+2)}$ is equal to:
- A) πi B) $\pi^2 i$ C) π D) π^2
10. If G is a group of even order $\forall a \neq e$. If $a^2 = e$ then G is:
- A) Abelian group B) Sub group
 C) Normal group D) Ring
11. The dominant mode in a rectangular wave guide is TE_0 because this mode has:
- A) No attenuation
 B) No cut off
 C) The highest cut off wavelength
 D) No magnetic field component
12. A positively charged particle moving along positive x direction enters an uniform electric field along positive y direction. The path of the particle inside the electric field is:
- A) Circular in a plane perpendicular to electric field.
 B) Circular in a plane parallel to electric field.
 C) Circular in a plane inclined to electric field.
 D) Parabolic
13. Suppose in a hypothetical space magnetic monopole exist. Then which of the following equation has to be modified?
- A) $\nabla \cdot D = \rho$ B) $\nabla \cdot B = 0$
 C) $\nabla \times E = -\frac{\partial B}{\partial t}$ D) All the above
14. The energy transported per unit area per unit time by an electromagnetic wave is expressed as:
- A) Energy flux B) Lorentz gauge
 C) Poynting vector D) Propagation vector
15. The power radiated at a frequency ω by an electric dipole is proportional to:
- A) ω B) ω^2 C) ω^3 D) ω^4

16. An electric charge 'q' is placed at the centre of a cube of side 'a'. The electric flux through any one face of the cube is:
- A) $\frac{q}{\epsilon_0}$ B) $\frac{q}{4\epsilon_0}$ C) $\frac{q}{6\epsilon_0}$ D) $\frac{q}{8\epsilon_0}$
17. The canonical momentum of a charged particle of mass m and charge q moving at a speed v along an electro-magnetic field of vector potential A is:
- A) $mv - \frac{q}{c}A$ B) $mv + \frac{q}{c}A$ C) $\frac{1}{2}mv + \frac{q}{c}A$ D) $\frac{1}{2}mv - \frac{q}{c}A$
18. Scleronomic constraints are:
- A) Independent of time
 B) Dependent of time explicitly
 C) Dependent of time implicitly
 D) None of the above
19. A meter scale is moving along its length at a speed $\sqrt{2}c$. Then its apparent length will be:
- A) $\frac{1}{\sqrt{2}}m$ B) $\frac{1}{2}m$ C) $\frac{\sqrt{3}}{2}m$ D) $\sqrt{\frac{3}{2}}m$
20. The expression $|\Psi_{(r,t)}|^2$ represents:
- A) Normalisation B) Position probability density
 C) Expectation value D) Time probability density
21. The minimum uncertainty in the velocity of electrons orbiting around the nucleus of radius r is:
- A) $\frac{h}{2\pi mr}$ B) $\frac{h}{2mr}$ C) $2hmr$ D) 0
22. The wave function in the ground state of hydrogen atom is given as $\Psi = Ae^{-r/a}$, where r is the distance from nucleus and a is constant. Then the value of A is:
- A) $\frac{1}{\sqrt{\pi a}}$ B) $\frac{1}{\sqrt{\pi} a}$ C) $\frac{1}{\sqrt{\pi a^3}}$ D) $\frac{1}{\sqrt{\pi a^5}}$
23. The wave function of a quantum particle is given by $\Psi_{(x)} = \frac{3}{5}\Phi_1(x) + \frac{4}{5}\Phi_2(x)$ where $\Phi_1(x)$ and $\Phi_2(x)$ are eigen function with energy eigen values $-1eV$ and $-2eV$ respectively. The energy of the particle in the state are $\Psi_{(x)}$ is:
- A) $-\frac{41}{25}eV$ B) $-\frac{11}{5}eV$ C) $-\frac{36}{25}eV$ D) $-\frac{7}{5}eV$

24. A particle of mass m is confined in two dimensional infinite square well potential of side a . The eigen energy of the particle in a given state is $E = \frac{25\pi^2\eta^2}{ma^2}$. The degeneracy of the state is----- degenerate.

A) 4 fold B) 3 fold C) 5 fold D) 2 fold

25. Four different one dimensional wave functions in a given interval is given below. Which among them have a physical significance in the interval?



A) I, III and IV B) I and III C) I and IV D) II and IV

26. A hydrogen atom remains in its ground state when its orbital electron is:
 A) At rest in the first orbit
 B) Orbiting in the first orbit
 C) Removed from the atom
 D) When light of wavelength $0.53\text{\AA}$ is absorbed from external source
27. If L_x represents x component of angular momentum and x, y and z represents x, y and z components of position operator, then the commutator $[L_x, y]$ is equal to:
 A) 0 B) $i\hbar x$ C) $i\hbar y$ D) $i\hbar z$
28. The Davisson and Germer experiment confirms:
 A) Particle nature of electron B) Particle nature of photon
 C) Wave nature of electron D) Wave nature of photon
29. In a scattering by a spherically symmetric potential, only p wave scattering occurs. The differential scattering cross section should have an angular dependence of the form:
 A) a^2 B) $a \cos^2 \theta$ C) $a \cos \theta$ D) $a + b \sin \theta$
30. The Clausius Clapeyron equation indicates that an increase in pressure increases the melting point, in the case of:
 A) All substances
 B) Substances that expand on solidification
 C) Substances that contract on solidification
 D) Substances that neither expand nor contract on solidification

31. Bose Einstein Condensation temperature T_B refers to the temperature below which:
- A) An assembly of Bose gas condenses to the liquid state
 - B) There is an appreciable occupation of the ground state in an electron system
 - C) There is a significant large occupancy of the ground state in a boson system
 - D) The bosons essentially behave like fermions.
32. The Fermi energy of a free electron gas depends on the electron density ρ as:
- A) $\rho^{-1/3}$
 - B) $\rho^{1/3}$
 - C) $\rho^{-2/3}$
 - D) $\rho^{2/3}$
33. In Fermi Dirac statistics the volume of phase cell is:
- A) h
 - B) h^2
 - C) h^3
 - D) $h^{2/3}$
34. Corresponding to the Fermi energy, at absolute zero the value of Fermi function for electron is:
- A) 1
 - B) 0
 - C) ∞
 - D) $\frac{1}{2}$
35. During an adiabatic expansion the internal energy of a gas:
- A) Remains constant
 - B) Increases
 - C) Decreases
 - D) Reduces to zero
36. In Stern Gerlach experiment the atomic beam whose angular momentum state is to be determined, must travel through:
- A) Homogeneous radio frequency magnetic field
 - B) Homogeneous static electric field
 - C) Inhomogeneous static magnetic field
 - D) Inhomogeneous radio frequency magnetic field.
37. The selection rule for orbital quantum number for electronic dipole transition in a single electron atom is:
- A) $\Delta l = 0, \pm 1$
 - B) $\Delta l = 0$
 - C) $\Delta l = \pm 1$
 - D) $\Delta l = 1$
38. Multiplicity of the state $^2D_{3/2}$:
- A) 1
 - B) 2
 - C) 3
 - D) 4
39. The Lande g factor for the level 3D_3 :
- A) $\frac{2}{3}$
 - B) $\frac{3}{2}$
 - C) $\frac{3}{4}$
 - D) $\frac{4}{3}$

40. The phenomenon used to study the energy levels of homonuclear molecule is--- effect.
A) Start B) Zeeman
C) Paschen Back D) Raman
41. Broadening of spectral lines cannot be explained by:
A) Doppler effect B) Uncertainty principle
C) Rayleighs criterion D) Pressure Variation
42. The time base of a CRO is developed by ---- waveform.
A) Sawtooth B) Square C) Triangular D) Sinusoidal
43. An increase in diode voltage leads to:
A) Increase in diode resistance
B) Decrease in diode resistance
C) No change in diode resistance
D) Increase or decrease depending upon nature of diode
44. In a transistor when the base emitter junction is open, the collector voltage is:
A) V_{cc} B) 0 V C) floating D) V_{Csat}
45. The JFET is:
A) Unipolar device B) Voltage control device
C) Current control device D) Both A and B
46. An SCR is turned off by:
A) Reducing anode voltage to zero
B) Reducing gate voltage to zero
C) Reverse biasing the gate
D) None of the above.
47. A 32-bit microprocessor has word length equal to ---- byte.
A) One B) Two C) Four D) Eight
48. The zero level detector is one application of:
A) Comparator B) Differentiator
C) Summing amplifier D) Diode
49. In large signal power amplifiers Class AB operation is often used. This is to:
A) Get maximum efficiency
B) Remove even harmonics
C) Overcome a cross over distortion
D) Reduce collector dissipation.
50. Which of the following Boolean statement is invalid?
A) $A + 1 = 1$ B) $A = \bar{A}$ C) $AA = A$ D) $A + 0 = A$

51. Number of flip-flops a modulo 12 counter must have:
 A) 12 B) 8 C) 4 D) 3
52. In a microprocessor the register which hold the address of the next instruction to be fetched:
 A) Accumulator B) Program counter
 C) Stack pointer D) Instruction register
53. Which of the following optical transducer is an active transducer?
 A) Photoemissive cell B) Photo diode
 C) Photo transistor D) Photo voltaic cell
54. A nucleus ruptures into two nuclear parts which have their velocities in the ratio 2:1. What will be the ratio of their nuclear size?
 A) $2^{1/3} : 1$ B) $1 : 2^{1/3}$ C) $3^{1/2} : 1$ D) $1 : 3^{1/3}$
55. The exchange of pions between two interacting nucleons is found in---- scattering.
 A) p-p B) e-p C) n-p D) n-n
56. The quadrupole moment of the nucleus is a:
 A) Scalar B) Vector C) Tensor D) None of these
57. According to the meson theory of nuclear forces a neutron -----.
 A) Emits a (π^-) meson and is converted into a proton
 B) Emits a (π^0) meson and is converted into a proton
 C) Emits a (π^+) meson and is converted into a proton
 D) Cannot be converted into a proton
58. The nucleus ^{41}Ca can be described by the single particle shell model- the single particle states occupied by the last proton and the last neutron, respectively are given by:
 A) $d_{5/2}$ and $f_{7/2}$ B) $d_{3/2}$ and $f_{5/2}$
 C) $d_{5/2}$ and $f_{5/2}$ D) $d_{3/2}$ and $f_{7/2}$
59. The reaction $e^+ + p^- \rightarrow \nu_e + \pi^-$ is forbidden because of Law of ---- conservation.
 A) Electron number B) Baryon number
 C) Momentum-energy D) Muon number
60. The quark structure of Δ^{++} is:
 A) uuu B) udu C) sss D) ddd

61. The weak interaction is responsible for:
 A) β -decay
 B) Neutron-neutron scattering
 C) Hadrons
 D) Heavy mesons
62. Isospin doublet is formed by:
 A) Electron and proton
 B) Electron and neutron
 C) Baryons
 D) Proton and neutron
63. Ice is an example of ---- system.
 A) Triclinic
 B) Hexagonal
 C) Orthorhombic
 D) Monoclinic
64. The co-ordination number in case of FCC is:
 A) 12
 B) 8
 C) 6
 D) 4
65. Point defect in a crystal constitutes of:
 A) Only vacancies in lattice sites
 B) Vacancies and impurity atoms
 C) Vacancies, impurities atoms and interstitials
 D) Impurities and interstitials
66. The number of independent elastic constants in an isotropic cubic solid:
 A) 1
 B) 2
 C) 3
 D) 4
67. If $\vec{g}$ is a reciprocal lattice vector, the Bragg's law can be written as:
 A) $\vec{k} + \vec{g} = 0$
 B) $2\vec{k} \cdot \vec{g} + g^2 = 0$
 C) $2\vec{k} \cdot \vec{g} + k^2 = 0$
 D) $2\vec{k} \cdot \vec{g} = 0$
68. In a non-dispersive medium with spacing 'a', the angular frequency ω and wave vector k are related to wave velocity as:
 A) $\omega = vk$
 B) $\omega = \frac{v}{k}$
 C) $\omega = \frac{2v}{a} \sin\left(\frac{ka}{2}\right)$
 D) $\omega = \frac{v}{a}$
69. The reciprocal lattice of a simple cubic lattice is:
 A) Monoclinic
 B) Triclinic
 C) Cubic
 D) Orthorhombic
70. The range of first Brillouin zone is:
 A) $\frac{\pi}{a} < k < \frac{2\pi}{a}$
 B) $-\frac{\pi}{a} < k < +\frac{\pi}{a}$
 C) $0 < k < \frac{2\pi}{a}$
 D) $-\frac{2\pi}{a} < k < +\frac{2\pi}{a}$

71. According to the Kronig Penny model, in the energy spectrum of electrons in solids, there are:
- Regular regions of only the allowed energies
 - Alternate regions of allowed energy and forbidden energy
 - Only the regular regions of forbidden energy
 - None of these
72. Vectors $\vec{A} = 2\hat{j} + 2\hat{k}$, $\vec{B} = 2\hat{i} + 3\hat{j} - 5\hat{k}$ and $\vec{C} = \hat{j} - \hat{k}$, the vector product $\vec{A} \times (\vec{B} \times \vec{C})$ is:
- In the same direction as $\vec{A}$
 - In the same direction as $\vec{B}$
 - In the same direction as $\vec{C}$
 - In the direction opposite to $\vec{C}$
73. What is the value of β for which $f(x,y) = 2x + 3(x^2 - y^2) + i(6xy + \beta y)$ is an analytic function of complex variable $z = x + iy$
- 1
 - 0
 - 3
 - 2
74. Divergence of a curl of a vector is always:
- 1
 - 0
 - infinity
 - not defined
75. Find the Laplace transform of the function $f(t) = t^2 e^{-2t}$
- $\frac{2}{(s+2)^3}$
 - $\frac{s}{(s+2)^3}$
 - $\frac{2}{(s+2)^2}$
 - $\frac{s}{(s+2)^2}$
76. Converting surface integral into volume integral is done by using:
- Gradient theorem for line integrals
 - Stoke's theorem
 - Green's theorem
 - Gauss's divergence theorem
77. What is the mean of first 29 natural numbers?
- 30
 - 15
 - 15.5
 - 14
78. Average energy of a Planck's oscillator is :
- $E = h\nu$
 - zero
 - $E = \frac{h\nu}{e^{\frac{h\nu}{kT}} - 1}$
 - $E = nh\nu$
79. Maxwell's thermodynamic relation is:
- $(\frac{\partial S}{\partial T})_T = (\frac{\partial P}{\partial V})_T$
 - $(\frac{\partial T}{\partial V})_S = -(\frac{\partial P}{\partial S})_V$
 - $(\frac{\partial T}{\partial P})_P = -(\frac{\partial V}{\partial P})_P$
 - None of these

80. Which of the following is true for a conservative force?
 A) which never do work
 B) work done in a closed path is zero
 C) work done depends on the path
 D) curl of the conservative force is a nonzero vector
81. Number of degrees of freedom a rigid body possess----.
 A) 3 B) 6 C) 9 D) infinite
82. The Lagrangian of a particle is given by the equation
 $L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r)$. The generalised momenta corresponding to r and θ are:
 A) $m\dot{r}$ and $mr^2\dot{\theta}$ B) $m\dot{r}$ and $m\dot{\theta}$
 C) $m\dot{r}^2$ and $mr^2\dot{\theta}$ D) $m\dot{r}^2$ and $mr^2\dot{\theta}^2$
83. A non-holonomic constraint may be expressed in the form of:
 A) inequality B) equality C) Vector D) None of these
84. The operator $i\frac{dy}{dx}$ is a:
 A) Hermitian operator B) Antihermitian operator
 C) unitary operator D) None of these
85. If x and p_x are the coordinate and momentum operators then $[x, p_x^n]$ is:
 A) $n\hbar p_x^n$ B) $n\hbar p_x^{n-1}$ C) 0 D) $n\hbar$
86. Energy eigen values of a quantum harmonic oscillator is:
 A) $n\hbar$ B) $(n + \frac{1}{2})\hbar$ C) $\hbar$ D) $n\hbar$
87. Locus of all points having same electric potential is called ----- surface.
 A) Charged B) Neutral
 C) Equipotential D) Magnetic
88. Electric field due to an infinitely long plane sheet of surface charge density σ is:
 A) $\frac{\sigma}{2\epsilon_0}$ B) $\frac{\sigma}{\epsilon_0}$ C) $\frac{\sigma^2}{2\epsilon_0}$ D) 0
89. Gauss' theorem of electric field E in differential form at vacuum:
 A) $\nabla \cdot E = \rho/\epsilon_0$ B) $\nabla \cdot E = 0$ C) $\oint E \cdot dA = 0$ D) $\oint E \cdot dA = Q_{total}$
90. Equation of continuity in magnetostatics takes the form (where J , ρ current density, charge density respectively)
 A) $\nabla \cdot J = -\frac{\partial \rho}{\partial t}$ B) $\nabla \times J = -\frac{\partial \rho}{\partial t}$ C) $\nabla \cdot J = 0$ D) $\nabla \cdot J = -\frac{\partial q}{\partial t}$

91. The magnetic vector potential **A** is related with magnetic field **B** as:
 A) $\nabla \times \mathbf{A} = \mathbf{B}$ B) $\nabla \cdot \mathbf{A} = \mathbf{B}$
 C) $\frac{\partial \mathbf{A}}{\partial t} = \mathbf{B}$ D) None of these
92. For an Op-amp with negative feedback, the output is:
 A) Fed back to the inverting input
 B) Decreased
 C) Fed back to the noninverting input
 D) Increased
93. A voltage follower:
 A) Has a voltage gain of unity
 B) Is noninverting
 C) Has no feedback resistor
 D) Has all of these
94. The number system which has a base 16:
 A) Binary B) Octal
 C) Hexadecimal D) Decimal
95. Convert $(312)_8$ into decimal:
 A) $(102)_{10}$ B) $(202)_{10}$ C) $(103)_{10}$ D) $(203)_{10}$
96. Which of these sets of logic gates are known as universal gates?
 A) NOR, NAND B) OR, NOT, XOR
 C) NOR, NAND, XOR D) NOR, NAND, OR
97. The 2's complement of 1011011 is:
 A) 0100010 B) 0110101 C) 0100001 D) 0100101
98. Which of the following parameter is non zero in lossless transmission line:
 A) Attenuation B) Resistance
 C) Phase constant D) Inductance
99. In a *P*-type semi-conductor, germanium is dopped with:
 A) Boron B) Aluminium C) Gallium D) All of these
100. A particle is scattered by a potential step, then the sum of reflection and transmission coefficient is:
 A) 0.5 B) 0 C) 2 D) 1
101. Choose the correct characteristic features of the nuclear force:
 A) Nuclear forces are short range forces and charge independent
 B) Nuclear forces are non-central forces
 C) Both A and B
 D) Neither A nor B

102. The electrons which contribute to isomer shift in Mossbauer spectroscopy are:
 A) s- electrons B) p- electrons
 C) d- electrons D) f- electrons
103. In Raman spectroscopy the radiation lies in the -----region of electro magnetic spectrum.
 A) Microwave radiation B) IR radiation
 C) Radio frequency radiation D) Visible
104. The relation connecting transition temperature T_c and critical magnetic field H_c of a super conductor is:
 A) $T_c = T_0[1 - (H_0/H_c)^2]$ B) $H_c = H_0(T_c - 1)$
 C) $H_c = H_0[1 - (T/T_c)^2]$ D) $H_c = H_0(T_c + 1)$
105. Lattice specific heat varies at lower temperature 'T' as:
 A) T B) T^3 C) $1/T^3$ D) $1/T$
106. Match the List I with List II
- | List I | List II |
|----------------------|------------------------|
| a. Metals | 1. Phonons |
| b. Super conductors | 2. Electrons and holes |
| c. Lattice vibration | 3. Cooper pairs |
| d. Semi-conductors | 4. Electrons |
- A) a-4, b-3, c-2, d-1 B) a-2, b-1, c-3, d-4
 C) a-4, b-3, c-1, d-2 D) a-2, b-1, c-4, d-3
107. The fermi level in an n-type semiconductor at 0 k lie:
 A) Half way between conduction band and donor level
 B) Below the donor level
 C) Coincides with intrinsic Fermi level
 D) None of these
108. The fermi level of sodium at 0K is 3 eV. The ratio of fermi level at 0 K with 1000K is:
 A) 0.93 B) 0.01 C) 1.93 D) 21
109. The number of atoms present in the unit cell of hcp structure is:
 A) 7 B) 4 C) 2 D) 6
110. The magnetisation of a material is related to its magnetic induction **B** and the field strength **H** by the relation:
 A) $M = (\frac{1}{\mu_0} \mathbf{B} - \mathbf{H})$ B) $\mathbf{H} = \mathbf{B} - \mu_0 \mathbf{M}$
 C) $\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M})$ D) $\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M})$

111. Diamagnetic material possesses:
 A) Induced dipole moment
 B) Permanent magnetic dipoles
 C) No permanent magnetic dipoles
 D) None of these
112. Curie- Weiss law is;
 A) $\chi_m = C/T$ B) $\chi_m = C/(T - \theta)$
 C) $\chi_m = C/\theta$ D) $\chi_m = C/(\theta - T)$
113. Which fissionable nuclear fuel occurs in nature?
 A) Plutonium B) Uranium C) Thorium D) Thallium
114. The reaction ${}^2_1D + {}^2_1D \rightarrow {}^4_2He + \pi^0$ cannot proceed via strong interactions because it violates the conservation of:
 A) Baryon number B) Electric charge
 C) Isospin D) Angular momentum
115. A linearly polarized wave is always:
 A) X- Polarized B) Transverse in nature
 C) A Longitudinal wave D) Y polarized
116. Which is the unique property of a LASER?
 A) Directional B) Coherence
 C) Amplitude D) Wavelength
117. The energy gap between the conduction band and the valence band in a good conductor is:
 A) Zero B) Wide C) Narrow D) Infinite
118. The plane parallel to y and z-axes of a crystal lattice has miller indices:
 A) (1 0 0) B) (0 1 0) C) (0 0 1) D) (1 1 1)
119. Micro canonical ensemble is a collection of essentially independent systems having:
 A) Same temperature, volume and no of particles.
 B) Same energy, volume and no of particles
 C) Same temperature, volume and chemical potential
 D) None of these
120. Photons obey the ----- statistics.
 A) Fermi-Dirac B) Bose-Einstein
 C) Maxwell Boltzmann D) None of these